

The Trip Economy: Which Foul-Drawing Skills Travel?

Introduction

A trip to the free-throw line is the most valuable scoring attempt in basketball. At league conversion, a two-shot trip yields 1.57 expected points, more than a shot from any zone on the floor. Yet foul drawing is measured coarsely. Free-throw rate treats all trips alike, and possession arithmetic approximates trip counts with the 0.44 coefficient. Recent work folded the free throws earned on fouled shots into shot value; the rest of the trip economy, and the skill behind it, remain unmeasured. We ask how players actually get to the line, what each channel of foul generation is worth, and which channels are repeatable player skills rather than circumstance.

Methods

We reconstruct every NBA free-throw trip, meaning the free throws awarded from a single non-technical foul, from official play-by-play across the three most recent seasons (2023-24 through 2025-26, 3,690 games). We trace each trip to the foul that caused it and classify the trips into eight classes in two tiers. Attempt-equivalent trips end the possession in place of a field-goal attempt (two- and three-shot shooting fouls, bonus). Add-on trips stack points on a possession that already stands (and-ones and rarer classes). Reconstruction is exact rather than estimated. Every player-game reconciles with the official box score and every player-season with an independent league source, yielding 88,347 trips across 1,723 player-seasons with zero discrepancies. The dataset and pipeline are public.

Results

The trip economy varies structurally across players. Among 284 players with 300 or more field-goal attempts in 2025-26, two-shot shooting-foul generation spans 3.1 to 14.0 trips per 100 attempts (10th to 90th percentile). Within that group, James Harden leads in three-shot fouls drawn (3.8 per 100), Nikola Jokić in off-ball bonus trips (7.0). A two-shot trip at a player's own conversion out-values his average field attempt for 279 of the 284 players; the median player gains +0.51 points per attempt. Channels persist differently year over year (422 player-season transitions across two season pairs, same ordering in both): two-shot shooting fouls $r = .87$, three-shot fouls $.76$, and-ones $.75$, bonus $.58$, residual classes $.31$. Within-season (split-half) reliabilities of $.81$ to $.95$ show the gradient is not measurement noise (Figure 1). The bonus gap is context: for players who changed teams, bonus persistence falls from $.635$ to $.420$ ($p = .006$), while every other channel travels intact (gaps at most $.04$, $p > .2$). The conventional 0.44 coefficient mismeasures the same players every year: its true value spans $.277$ to $.481$, persists at $r = .53$, and moves True Shooting by up to 1.5 percentage points (Table 1).

Conclusion

Drawing shooting fouls and finishing and-ones are player skills that travel. The off-ball bonus trip partly belongs to the team. The split is directly actionable. Shooting-foul generation projects across team changes, bonus volume deserves a discount in trades and free agency, and exact trip accounting removes a persistent, player-shaped error from efficiency metrics.

Foul-drawing channels persist differentially, and the off-ball channel partly belongs to the team

Each dot: the correlation between a player's rate in one season and the next, computed separately over player-season pairs where he stayed with his team (blue, $n = 306$) and where he changed teams (orange, $n = 116$); p-values test each blue-orange gap.

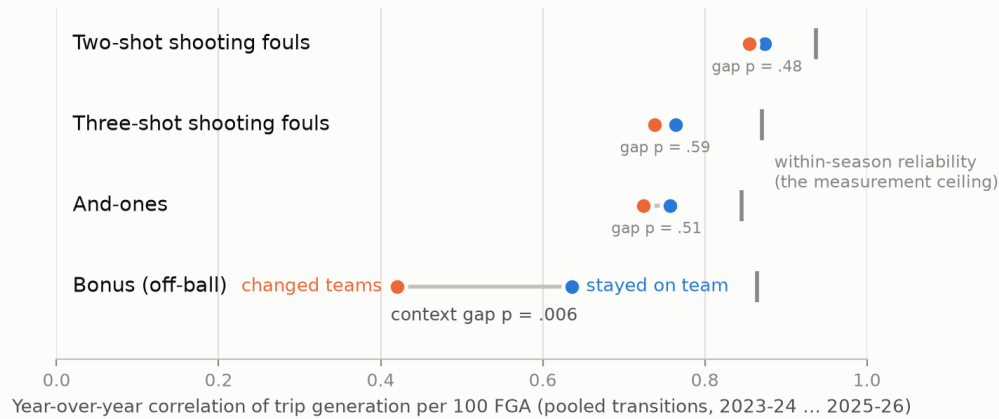


Figure 1. The persistence gradient and the context test, pooled over 2023-24 ... 2025-26.

Table 1. The trip taxonomy, measured. Volume and rate columns are 2025-26 (players with ≥ 300 field-goal attempts); both stability columns are pooled over 2023-24 ... 2025-26.

Channel	Tier	League trips (2025-26)	Median /100 FGA (p10–p90)	Year-over-year r	Split-half reliability
Two-shot shooting fouls	attempt-equivalent	18,846	8.1 (3.1–14.0)	0.87	0.94
Three-shot shooting fouls	attempt-equivalent	933	0.2 (0.0–1.1)	0.76	0.87
And-ones	add-on	5,860	2.6 (0.9–4.5)	0.75	0.85
Bonus (off-ball)	attempt-equivalent	4,760	1.8 (0.8–3.7)	0.58	0.86
Other add-on (flagrant, away-from-play, transition, clear path)	add-on	549	—	0.31	—
All trips	—	30,948	—	0.86	0.94

Note. The true attempt-equivalent coefficient — the exact replacement for the conventional 0.44 — has median 0.427 across qualified players (p10–p90 0.385–0.455, full span 0.277–0.481) and persists year over year at $r = 0.53$.